Branch-Dependent Proper Time

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Abstract

We attach the Károlyházy uncertainty to *proper time* and postulate that its fluctuations are independent across quantum branches. Interference is sensitive to the difference of branch phase contributions. A fully distinguishable which-path carrier contributes its rest energy mc^2; internal branches contribute the transition energy. For a centre-of-mass interferometer our model predicts a reduction of the fringe visibility to $V/V_0 = \exp(-\chi)$ with $\chi = (mc^2/\hbar)^2(\xi\, t_P^2\, t)^{2/3}$, where t is the superposition time, t_P is the Planck time, and $\xi = \mathcal{O}(1)$ is a dimensionless coefficient in the Károlyházy relation. Attributing the entire contrast deficit in the Vienna sodium-nanoparticle interferometry at $m \approx 172$ kDa and $v \approx 160$ m/s to the mechanism gives $\xi \lesssim 1.1$. A decisive test is near: at $m \sim 1$ MDa and $v \approx 25$ m/s the model predicts practically complete loss of interference, whereas the standard quantum model expects measurable fringes with ordinary contrast.

Keywords: proper time; branch clock independence; Károlyházy uncertainty; decoherence; matter-wave interferometry; quantum gravity

1 Introduction

In quantum mechanics the status of the wave function—an ontic object or a computational bookkeeping of an ensemble of possibilities—remains unsettled. We adopt the second, ensemble reading [1, 2]: $|\psi\rangle = \sum_i c_i|i\rangle$ catalogues distinct variants of the system, and $|c_i|^2$ are their weights.

In the relational view of time, a clock is a physical subsystem and "time" is a correlation of the clock and the rest of the system within their joint state [3]. If each branch is a genuine variant of the system, then it carries its own world line and hence its own proper time. Penrose reaches a structurally identical statement from the opposite direction: each component of the superposition sources its own space-time, and "since there is no canonical way of relating the time parameters in the two space-times, a fundamental uncertainty in the time-translation operator arises" [4]. Consistently with this, proper time accumulates along a world line and is not fixed by a single common scale for distinct four-dimensional paths: in an interferometer with different trajectories one cannot, without an additional postulate, impose a single common proper-time parameter on all branches.

The idea that the inaccuracy of a physical clock can extinguish interference is older than the present model. Salecker and Wigner bounded the accuracy of a clock by its mass [5]. Károlyházy, and later Ng and van Dam, showed that gravity stops one from sending this bound to zero by making the clock arbitrarily heavy: what remains is the cubic relation $\delta t^3 \sim t_P^2 t$ [6, 7, 8]. Gambini, Porto and Pullin turned that relation into a decoherence law: in the energy basis of a single Hamiltonian the off-diagonal element decays as $\exp(-\omega_{nm}^2 t_P^{4/3} t^{2/3})$, where ω_{nm} is a difference of eigenfrequencies [9, 10]. While states of equal energy interfere, $\omega_{nm} = 0$ and there is no decay. Milburn's intrinsic decoherence belongs with that decay; Adler's simplest equation coincides with it [11, 12, 13]. There too the populations are conserved and the dephasing follows an energy difference, but the exponent is linear in time rather than growing as $t^{2/3}$.

Alongside these are constructions in which time is attached to a place rather than to a branch. Diósi coupled rest energy to random shifts of local time [13]. With his spatial correlator the speed of light cancels, and the master equation coincides with Diósi–Penrose collapse: the rate is set by the gravitational self-energy rather than by mc^2, and population redistribution and spontaneous emission appear [14]. In quantum-clock interference a proper-time difference along two paths records which-path information unitarily, through internal degrees of freedom or a gravitational gradient [15, 16]; there is no stochastic Planckian noise.

The present work differs in that it takes the cubic clock bound and assigns independent proper-time shifts to the branches of one split. In place of the eigenfrequency difference ω_{nm} stands the energy of whatever is in superposition: for which-path this is mc^2. Arms of equal total energy therefore still lose their mutual phase. A detailed comparison of the channels is given in § 8.

2 Definitions and the postulate

2.1 Notation.

- t — the time the system spends in superposition (a proper-time interval along the world line);
- δt — the proper-time uncertainty over the interval t; δt_a — the realized shift on branch a (the postulate), $\mathrm{Var}(\delta t_a) = \delta t^2$;
- $t_P = \sqrt{\hbar G/c^5}$ — the Planck time;
- E_{br} — the characteristic energy of the *split* subsystem in the interfering branch pair;
- $E_{a|b}$ — the energy of branch a in the superposition: the energy carried on that branch by degrees of freedom that distinguish it from branch b, measured from the physical vacuum or ground state of the split sector after removing the sector common to both branches;
- $\omega_{\mathrm{eff}} = E_{\mathrm{br}}/\hbar$ — the corresponding effective phase-noise frequency;
- m — the carrier mass in the which-path case (then $E_{\mathrm{br}} = mc^2$, $\omega_{\mathrm{eff}} = \omega_C = mc^2/\hbar$);
- $\xi = \mathcal{O}(1)$ — a dimensionless numerical coefficient.

2.2 Definition of a branch.

A branch a is the term $c_a|a\rangle$ of one superposition $|\psi\rangle = \sum_a c_a|a\rangle$, differing in a *physical degree of freedom* that participates in interference: a spatial-motional state (which-path or current), or an eigenstate of the internal Hamiltonian H_{int} (spin, a hyperfine or optical level, a qubit). One connected and indistinguishable wave packet carries one clock. Branching occurs when evolution splits the state into distinguishable terms, and ceases when those terms recombine or the apparatus registers which-path. Which terms of this superposition count as branches is fixed by one split: the split whose interference is read out. In which-path the arms are fixed by the preparation and by the interferometer. Rewriting the same vector in another basis does not create new clocks (see § 9.1).

2.3 Energy of a branch.

The energy of branch a in the superposition, $E_{a|b}$, is the energy carried on that branch by the degrees of freedom that distinguish it from branch b. The zero of energy is the physical vacuum or the ground state of the split sector. A sector represented identically on both branches is common: it is subtracted before the remaining energy is multiplied by δt_a and δt_b. What

couples to the independent clock shifts is this $E_{a|b}$. The energies $E_{a|b}$ and $E_{b|a}$ are fixed without the probabilistic weights $|c_a|^2$ and $|c_b|^2$: they are the energies of the degrees of freedom that distinguish the branches, not the mean energy of the superposition.

On a motional pair both branches carry the same exclusive rest energy, $E^{\mathrm{mot}}_{a|b} = E^{\mathrm{mot}}_{b|a} = E^{\mathrm{mot}}_{\mathrm{br}}$. When the spatial packets of the branches are distinguishable, $E^{\mathrm{mot}}_{a|b} = mc^2$. How that energy is assembled from the particle masses is given in § 2.4.

For internal states $|0\rangle, |1\rangle$ of a common carrier the zero lies at the ground state: branch $|0\rangle$ carries no exclusive energy, and branch $|1\rangle$ carries the transition energy, $E^{\mathrm{int}}_{0|1} = 0$ and $E^{\mathrm{int}}_{1|0} = \Delta E$. If motion and an internal level both differ, the two contributions add on each branch: $E_{a|b} = E^{\mathrm{mot}}_{\mathrm{br}} + E^{\mathrm{int}}_{a}$ and $E_{b|a} = E^{\mathrm{mot}}_{\mathrm{br}} + E^{\mathrm{int}}_{b}$.

A global shift $H \mapsto H + C\mathbf{1}$ adds the same number to the reference energy of the common sector or of the ground state. After that sector is subtracted, $E_{a|b}$ and $E_{b|a}$ are unchanged, and so are the visibilities.

2.4 Assembly from the particle masses.

Let $E^{\mathrm{mot}}_{\mathrm{br}}$ be the rest energy of those particles by which the already given branches differ. The full mass of the body enters only when the spatial packets of all its particles are distinguishable between the two branches. For particles of species s—one type and one mass m_s—let $\gamma^{(1)}_{a,s}$ and $\gamma^{(1)}_{b,s}$ be the one-particle reduced density matrices of the branches, normalized by $\operatorname{Tr}\gamma^{(1)}_{a,s} = \operatorname{Tr}\gamma^{(1)}_{b,s} = N_s$. From these matrices the rest energy of a motional pair is assembled from the number of distinguishable particles of each species:

$$\mathcal{D}_s(a,b) = \frac{1}{2}\left\|\gamma^{(1)}_{a,s} - \gamma^{(1)}_{b,s}\right\|_1, \qquad E^{\mathrm{mot}}_{\mathrm{br}} = c^2 \sum_s m_s\, \mathcal{D}_s(a,b). \tag{1}$$

Identical particles of one species share one matrix. Different species are added separately: each has its own mass and its own packet overlap. The definition does not depend on the one-particle basis and applies to identical particles [17]; throughout, $0 \le \mathcal{D}_s \le N_s$. A state that is the same on both branches does not enter the difference of the matrices. For a motional pair this energy is the exclusive rest energy of each branch,

$$E^{\mathrm{mot}}_{a|b} = E^{\mathrm{mot}}_{b|a} = E^{\mathrm{mot}}_{\mathrm{br}} = c^2 \sum_s m_s \mathcal{D}_s(a,b). \tag{2}$$

This is the energy, relative to the vacuum, of the occupied branch-exclusive matter modes. The formula does not choose the decomposition into branches. The accounting through $E_{a|b}$ and $\mathcal{D}_s$ is a proposed rule for the extensions. Pairwise equality $E_{a|b} = E_{b|a}$ does not by itself give consistent noise for three or more branches: a branch must carry one energy that does not depend on which other branch it is compared with, or else the relative phases may fail to close a cycle (see § 9.2).

2.5 Proper-time fluctuation.

The proper-time fluctuation δt is adopted from the Károlyházy uncertainty relation and is not a new postulate of this work.

The accumulated uncertainty of the proper-time interval t measured along a world line is

$$\boxed{\delta t^3 = \xi\, t_P^2\, t,} \tag{3}$$

with $\xi = \mathcal{O}(1)$. Equation (3) is the temporal form of the Károlyházy uncertainty relation [6]; the same cubic scaling is consistent with the clock arguments of Salecker–Wigner [5], Ng–van Dam [7, 8], and the holographic bound on space-time uncertainty [18]. We do not fix the

numerical coefficient ξ: in these works it is likewise never pinned down more precisely than to $\mathcal{O}(1)$.

Equation (3) fixes only the variance of a single interval t: the law $\mathrm{Var}[\delta t(t)] \propto t^{2/3}$ does not by itself determine the full stochastic process, and the covariance of successive intervals, the noise spectrum, and the spin-echo response are not fixed by it. If Gaussianity of the process, self-similarity, and stationary increments are additionally imposed, the natural completion is fractional Brownian motion with Hurst exponent $H = 1/3$ [19]. In that completion, disjoint increments are anticorrelated, the low-frequency spectrum of fractional Gaussian noise behaves as $S(f) \propto |f|^{1/3}$, and echo requires an explicit calculation with the corresponding filter function rather than an automatic cancellation of a quasistatic phase. These spectral properties follow from an additional temporal postulate, not from the relation (3).

We stress where gravity enters δt. The Salecker–Wigner lower bound for a clock of mass m_{cl}, $\delta t \gtrsim (\hbar t/m_{\mathrm{cl}}c^2)^{1/2}$, vanishes as $m_{\mathrm{cl}} \to \infty$. The "not a black hole" constraint stops that limit: the optimal m_{cl} drops out and leaves the mass-independent minimum $\delta t^3 \sim t_P^2 t$ adopted in Equation (3). The rest mass is equally essential: the uncertainty δt is defined along a world line with nonzero proper time; without rest mass there is no proper time. For a photon $m = 0$, $\mathrm{d}\tau = 0$, and the rest-energy channel does not apply to a light-like carrier.

2.6 Postulate: branch clock independence.

This is the postulate of the model. In standard quantum mechanics one time parameter is common to the whole superposition. Here branches a and b have their own proper-time shifts, δt_a and δt_b, and those shifts are independent.

For two branches a, b (in the sense of the definitions above) over the superposition interval t (the same as in (3)), the realized shifts $\delta t_a, \delta t_b$ are independent, centered Gaussian random variables of accumulated proper time with $\mathrm{Var}(\delta t_a) = \mathrm{Var}(\delta t_b) = \delta t^2$ and $\mathrm{Cov}(\delta t_a, \delta t_b) = 0$. The variances are always equal. Branches of one split separate at one moment and, until they recombine or the apparatus reads which-path, share one interval t, so (3) assigns them the same δt. This is a substantive postulate, motivated by the ensemble ontology (§ 1): distinct variants do not share a common clock; there is no strict derivation from that ontology, and what remains irreducible is the premise that the holographic information budget is allocated *per branch* rather than as a single shared register.

Within one branch, all degrees of freedom *not* participating in the split share that branch's clock. They do not form a branch of their own: the part that is the same on both arms enters each term of the split, advancing with δt_a on term a and with δt_b on term b. The energy of this common sector does not enter the relative phase: it is subtracted before multiplication by the shifts (§ 2.3).

Crucially, δt_a is not a fluctuation of an external metric field and not environmental noise common to nearby trajectories. It is an *ontic* random shift of proper time; its magnitude over the interval t is set by δt from (3)—the holographic estimate of such uncertainty [18]. An environmental source—for example, a difference of gravitational potentials on the interferometer arms—would give a *correlated* shift δt_a, the closer to common the nearer the branches, and it would cancel in the relative phase; but that is a systematic shift, not noise. We take the correlation of the branch noises to be appreciable only for a separation smaller than $\ell = c\delta t$. At a larger separation the shifts are independent. This is an additional assumption: it does not follow from the relation (3) (see § 9.4). For resolved arms of a laboratory interferometer the separation is much larger than ℓ, so the proximity of such arms does not correlate their clocks. The holographic uncertainty remains attached to the split subsystem of a branch, not to spatial points. Attachment to discrete branches rather than positions is essential: noise independent for arbitrarily close *positions* would mean a random phase field with unbounded gradient—hence a force and momentum diffusion. Noise independent for discrete *branches* is pure dephasing without force.

3 Dephasing mechanism

3.1 Relative phase and visibility.

The phase contribution of the split subsystem gives the relative phase

$$\Delta\varphi = \frac{E_{a|b}}{\hbar}\,\delta t_a - \frac{E_{b|a}}{\hbar}\,\delta t_b, \tag{4}$$

Interference involves the *difference* of branch-exclusive contributions, not an absolute eigenvalue of the total Hamiltonian.

By the postulate, $\mathrm{Var}(\delta t_a) = \mathrm{Var}(\delta t_b) = \delta t^2$; then:

$$\mathrm{Var}(\Delta\varphi) = \frac{E_{a|b}^2 + E_{b|a}^2}{\hbar^2}\,\delta t^2. \tag{5}$$

If the centre of mass branches (which-path), we use the decomposition $H = mc^2 + P^2/(2m) + H_{\mathrm{int}}$ for a composite system: rest energy, centre-of-mass kinetic energy, and the Hamiltonian of the internal degrees of freedom. By § 2.3, $E_{a|b} = E_{b|a} = E_{\mathrm{br}} = mc^2$, and (5) gives

$$\mathrm{Var}(\Delta\varphi) = 2\,\omega_{\mathrm{eff}}^2\,\delta t^2, \qquad \omega_{\mathrm{eff}} = \frac{mc^2}{\hbar}. \tag{6}$$

Since $\delta t_a, \delta t_b$ are Gaussian by the postulate, the relative phase $\Delta\varphi$ is also Gaussian and $\langle e^{i\Delta\varphi}\rangle = \exp[-\mathrm{Var}(\Delta\varphi)/2]$; the factor 2 from the two branches and the factor $1/2$ from Gaussian averaging cancel. In the which-path case, for fringe visibility (V relative to the reference V_0):

$$\boxed{\frac{V}{V_0} = e^{-\chi}, \qquad \chi = \omega_{\mathrm{eff}}^2\,\delta t^2 = \left(\frac{mc^2}{\hbar}\right)^2 \left(\xi\, t_P^2\, t\right)^{2/3}.} \tag{7}$$

This is the ratio of visibilities when the dephased contribution is unmodulated and the mean level used to normalize the contrast is preserved. For a given subsequent measurement the channel gives the mixture of signals $S_q = qS_{\mathrm{coh}} + (1-q)S_{\mathrm{deph}}$. For one pair of orthogonal branches

$$\boxed{\chi_{ab} = \frac{E_{a|b}^2 + E_{b|a}^2}{2\hbar^2}\,\delta t^2.} \tag{8}$$

For independent proper-time shifts the variances of the phase contributions add, so (8) contains the sum of squares $E_{a|b}^2 + E_{b|a}^2$, not the square of the sum of the energies. An overall unweighted sum of the energies of all branches is not introduced in this work as a property of the superposition.

For internal branches $|0\rangle, |1\rangle$ with a common carrier $|C\rangle$ we take $E_{\mathrm{br}} = \Delta E$ and $\omega_{\mathrm{eff}} = \Delta E/\hbar$; since here $E_{0|1} = 0$ and $E_{1|0} = \Delta E$, dephasing is twice as weak as in the symmetric case: $\chi = \omega_{\mathrm{eff}}^2\,\delta t^2/2$.

When every arm carries the same energy, each arm's own δt_a gives one factor $e^{-\chi}$ to every arm pair of one split, including three or more arms. The scaling form $\chi \propto m^2 t^{2/3}$ in a centre-of-mass interferometer follows from the relation (3) and the postulate and constitutes the central testable prediction; the overall coefficient carries an $\mathcal{O}(1)$ uncertainty (through $\xi^{2/3}$ and the interpretation of δt; see § 7). The mass-quadratic dependence follows from rest-phase accumulation ($\omega_{\mathrm{eff}} \propto mc^2$) and matches the typical dephasing scaling in models of metric and conformal space-time fluctuations [20, 21]; in the high-mass interferometry programme, this mass growth of the signal is highlighted as a practical motivation for searching for hypothetical effects [22].

Using $\omega_C = mc^2/\hbar$ requires a caveat: in standard quantum mechanics the Compton phase cancels in a closed interferometer [23], whereas atom interferometry has been interpreted as

providing access to the mass frequency [24]. The present model adds the independent premise that the vacuum-referenced rest energy of branch-exclusive massive modes couples to the independent shifts $\delta t_a, \delta t_b$ and thereby becomes a random *relative* contribution. It does not couple an arbitrary identity term in the total Hamiltonian: the definitions first remove the common sector and its reference energy.

3.2 Dephasing map.

For fully resolved arms the spatial packets of every constituent are distinguishable, $\mathcal{D}_s = N_s$, and the energy of each arm is Mc^2. The arms are given by pairwise orthogonal projectors P_a that sum to the identity on the space of this split. The physical protocol fixes these projectors: they do not depend on the input density matrix or on a later choice of readout.

While the arms propagate separately, each is assigned one phase

$$\varphi_a = \frac{Mc^2}{\hbar}\,\delta t_a, \qquad \Delta\varphi_{ab} = \varphi_a - \varphi_b. \tag{9}$$

For two arms there is one relative phase, $\Delta\varphi_{ab} = \varphi_a - \varphi_b$. For three or more arms the same differences close a cycle: for any three arms a, b, c the sum $\Delta\varphi_{ab} + \Delta\varphi_{bc} + \Delta\varphi_{ca} = 0$ holds identically, because each arm has one phase. Independence of the shifts δt_a is an assumption of the postulate, not a consequence of orthogonality. Averaging over these Gaussian shifts describes the noise after the ordinary propagation that preserves the arm labels, and gives, on this interval, the dephasing map

$$\Phi_t(\rho) = q\rho + (1-q)\sum_a P_a \rho P_a, \qquad q = \exp\left[-\left(\frac{Mc^2}{\hbar}\right)^2 (\xi\, t_P^2\, t)^{2/3}\right]. \tag{10}$$

Here M is the mass of such an arm, and m is the which-path carrier mass in (7). For $M = m$ the factor q coincides with $e^{-\chi}$. The trace is preserved. The diagonal blocks of the arms, including coherences inside an arm, are unchanged, and coherences between different arms are multiplied by q. For $0 \le q \le 1$ the map is a mixture of the identity channel and complete erasure of coherences in the arm basis, so it is completely positive. This is the dynamics of one interval of separate propagation. There is no general theory here of repeated splittings and recombinations (see § 9.3).

4 Planck mass boundary

The decoherence time t_{dec} is defined by the condition $\chi = 1$. From (7):

$$t_{\mathrm{dec}} = \frac{t_C^3}{\xi\, t_P^2}, \qquad t_C = \frac{\hbar}{mc^2}. \tag{11}$$

Setting $t_{\mathrm{dec}} = t_C$, we obtain $t_C = \sqrt{\xi}\, t_P$, i.e. the characteristic scale

$$m_* = \frac{m_P}{\sqrt{\xi}}, \qquad m_P = \sqrt{\hbar c/G} \approx 2.2 \times 10^{-8}\ \mathrm{kg}. \tag{12}$$

For $\xi = \mathcal{O}(1)$ this is of the order of the Planck mass. For $m \gg m_*$ we have $t_{\mathrm{dec}} \ll t_C$: inter-branch coherence vanishes over an interval below the uncertainty δt from (3) and in principle inaccessible to direct measurement.

5 Growth of the von Neumann entropy

Dephasing changes the von Neumann entropy of the averaged state. For a fixed realization of the shifts $\delta t_a, \delta t_b$ the evolution is unitary: the vector

$$|\psi\rangle = c_a|a\rangle + c_b e^{i\Delta\varphi}|b\rangle \tag{13}$$

remains pure, $S = 0$; the phase $\Delta\varphi$ is given by (4). A mixture appears after averaging over the distribution of these shifts. Since $\delta t_a, \delta t_b$ are Gaussian by the postulate, the off-diagonal element is multiplied by $\langle e^{i\Delta\varphi}\rangle = e^{-\chi}$:

$$\overline{\rho}_{ab} = \rho_{ab}(0)\, e^{-\chi}, \tag{14}$$

where χ is the exponent (8). The von Neumann entropy refers to this matrix,

$$S = -\operatorname{Tr}(\overline{\rho}\ln\overline{\rho}). \tag{15}$$

For two branches of equal weight the eigenvalues of $\overline{\rho}$ are

$$\lambda_\pm = \frac{1 \pm e^{-\chi}}{2}, \qquad S = -\lambda_+ \ln\lambda_+ - \lambda_- \ln\lambda_-. \tag{16}$$

At $\chi = 0$ the spectrum is $(1, 0)$ and $S = 0$. Since $\chi \propto t^{2/3}$, as the superposition time grows λ_+ and λ_- approach each other and S increases. As $\chi \to \infty$ the coherence has vanished, the spectrum coincides with the branch weights, and the entropy saturates at the Shannon value

$$S \to -\sum_i |c_i|^2 \ln|c_i|^2 \tag{17}$$

(for n equal weights, $S \to \ln n$). A further increase of χ adds no new eigenvalues.

The populations $|c_i|^2$ stay constant throughout. λ_+ decreases from 1 to 1/2 and λ_- increases from 0 to 1/2: the off-diagonal elements decay as $e^{-\chi}$. For equal weights the original vector $|+\rangle = (|a\rangle + |b\rangle)/\sqrt{2}$ remains an eigenvector of $\overline{\rho}$, with eigenvalue $\lambda_+ = (1 + e^{-\chi})/2$. This is the growth of the entropy of the averaged state of the split system itself, not of a reduced subsystem. In environmental decoherence the joint state of system and environment can remain pure, and entropy appears after the environment is traced out. Here there is no environment recording which-path information. The mixture and the growth of S arise by averaging over the unknown phases. The averaged dephasing is non-unitary, yet linear in the density matrix, $\rho_{ab} \mapsto \rho_{ab} e^{-\chi}$. The growth of S refers to $\overline{\rho}$ and does not contradict the unitarity of a single realization.

6 Examples

6.1 Full resolution.

This is the case of a rigid cluster whose branches are centre-of-mass shifts of the whole body, with no change of the internal configuration. An example is a sodium cluster in a matter-wave interferometer: two spatially resolved arms of one and the same body. When the single-atom packets of these arms do not overlap, $\mathcal{D}_{\mathrm{Na}} = N$ and $E_{\mathrm{br}}^{\mathrm{mot}} = N m_{\mathrm{at}} c^2 = Mc^2$: the energy that enters is the mass of the whole cluster, and the map of § 3.2 applies. For the Vienna cluster this is $m \approx 172$ kDa; the comparison with the measured visibility is in § 7.

6.2 NOON state.

This is the full resolution of § 6.1 for N atoms of one split, not the mass of a rigid cluster. A NOON state

$$(|N,0\rangle + |0,N\rangle)/\sqrt{2} \tag{18}$$

is one split: two collective branches with $\mathcal{D} = N$ and $E_{\rm br} = Nmc^2$. A single clock for the collective branch is an additional assumption. Under that assumption the map (10) applies with the replacement $M \to Nm$. For $N = 10^5$ ^{87}Rb atoms and $t = 10$ ms, Equation (7) predicts $\chi \simeq 1.4 \times 10^3$ at $\xi = 1$, i.e. practically complete suppression of interference for a massive motional NOON state. This additional falsifiable prediction does not yet conflict with experiment: demonstrated atomic NOON states remain few-particle; for example, Ref. [25] realized a second-order state.

6.3 Incomplete resolution.

If the packets of one species have separated only in part, $0 < \mathcal{D}_s < N_s$, and that species contributes $m_s\mathcal{D}_s c^2$ rather than $m_s N_s c^2$. An example is the same two arms of a cluster while their separation is comparable to the packet width: some of the atoms still overlap, and $\mathcal{D}_{\rm Na}$ grows from zero toward N as the arms separate. Species are not mixed in the sum. In a body of several species the nuclei may already be distinguishable while the electron packets still overlap; the terms $m_s\mathcal{D}_s c^2$ then differ and are added separately. This is a sketch. The pair energy here depends on the partner, and the packets are not given as orthogonal projectors. What is missing is a physical specification of the branches, one joint noise, and one overall map; their consistency does not follow from the map (10) (see § 9.2).

6.4 Common carrier.

Only an internal level differs, and the spatial-motional part $|C\rangle$ is common. The rest mass of the carrier does not enter the difference of the matrices; $E^{\rm int}_{\rm br} = |\Delta E|$, with the ordered values $E^{\rm int}_{0|1} = 0$ and $E^{\rm int}_{1|0} = \Delta E$ from the definitions. An example is an atomic clock: a superposition of two internal levels of one atom, both on the same trajectory $|C\rangle$. The arms differ by the level, not by position. For a microwave transition $\Delta E \ll mc^2$, and χ is negligible. For these two levels there is a single relative phase, the ground-state phase vanishes, and averaging produces a channel of the same form as (10); the dephasing is again twice as weak as in the symmetric case.

6.5 Collective mode.

A mode of a common body is displaced by much less than the atomic zero-point motion. $E_{\rm br}$ receives the mode energy, not $\mathcal{D}_s mc^2$. An example is a vibrational mode of a molecule, or a phonon in a solid, at such a displacement. The atoms of the two branches are almost indistinguishable by position, so a count through $\mathcal{D}_s mc^2$ would insert the rest energy of distinguishable atoms, which this split does not have. This is a sketch: an orthogonal basis of the mode is not given, and there is no single joint noise with one overall evolution map. Consistency with the map (10) does not follow (see § 9.2).

6.6 Current.

The current in the definitions is the persistent current in the superconducting loop of a flux qubit. The branches are the states with opposite sense of that current, clockwise and counterclockwise. The loop is the same in both, so the condensate electrons of the two branches overlap almost completely: the energy receives $\mathcal{D}_e$, not the mass of every electron. A microscopic analysis of the BCS states gives $\mathcal{D}_e \simeq 42$–5750 for the devices considered, against 10^6–10^{10}

condensate electrons [26, 27]. The naive count puts the rest energy of the entire condensate into $E_{\rm br}$, as in full resolution $\mathcal{D}_e = N$. The upper value of 10^{10} electrons then gives $E_{\rm br} \simeq 8.2\times10^{-4}$ J and, for $t = 100$ μs and $\xi = 1$, $\chi \simeq 26.5$; the observed coherence of current states would exclude the model. With $\mathcal{D}_e$ at the same t one obtains $\chi \lesssim 9\times10^{-12}$: existing flux qubits are safe for the model. The difference between the naive and the microscopic count is not an $\mathcal{O}(1)$ uncertainty. The two senses of the current give a two-arm channel of the same form as (10), with the energy assembled from $\mathcal{D}_e$.

6.7 Phase noise and the qubit register.

The definitions and the postulate act on any superposition, including a computational one. For strictly internal levels $|0\rangle, |1\rangle$ with a common one-particle spatial-motional density matrix, the definitions give $E_{\rm br} = \hbar\omega_q = \Delta E$; the mass of the chip and resonator enters the common $|C\rangle$ and does not enter $\omega_{\rm eff}$. This statement cannot automatically be transferred to a superposition of opposite persistent currents, for which Equation (1) applies as discussed in § 6.6. Superconducting resonators storing a single photon for $\sim$34 ms with $\gtrsim 10^{23}$ electrons in their walls [28] illustrate the internal case: with $E_{\rm br} = \Delta E$ of the photon, not the rest energy of the common walls, long-lived coherence is consistent with the model. The random relative phase between a pair of internal branches over time t:

$$\sigma(t) = \frac{\sqrt{E_{a|b}^2 + E_{b|a}^2}}{\hbar}\,\delta t = \omega_q(\xi\, t_P^2\, t)^{1/3} \quad (E_{0|1} = 0,\ E_{1|0} = \hbar\omega_q). \tag{19}$$

For a superconducting qubit ($\omega_q \approx 2\pi \times 5$ GHz) and $t = 1$ s: $\delta t \approx 1.4 \times 10^{-29}$ s, $\sigma \approx 4.5 \times 10^{-19}$ rad (the corresponding dephasing parameter $\chi = \sigma^2/2$). The variance of one fixed pair does not depend on the number of the other branches, but it does depend on the energies of that pair, $(E_a^2 + E_b^2)\delta t^2/\hbar^2$. One and the same σ therefore does not apply to all 2^n branches. A change of basis defines a different split, and the map (10) does not generate it (see § 9.1). If d excitations form one collective branch against the ground state, $E = d\hbar\omega_q$ and $\chi \propto d^2$; for independently split qubits the exponents add, $\chi \propto d$. For the collective pair "all excited / all in the ground state" the conditional estimate $n\,\omega_q\,\delta t \lesssim 1$ gives $n \lesssim 10^{18}$—far beyond practical horizons. This is not a universal limit on the register.

6.8 Product state.

A product state

$$\Big[(|L\rangle + |R\rangle)/\sqrt{2}\Big]^{\otimes N} \tag{20}$$

contains N one-atom superpositions. If the physical phase noises of the atoms are independent, each is its own split. Observed first-order coherence compares components with $\mathcal{D} = 1$ and acquires the atomic energy mc^2. Under the same independence the joint map is the tensor product of one-atom channels of the form (10) with energy mc^2.

7 Experimental test

7.1 Matter-wave interferometry: the primary probe.

By the definitions and the postulate (§ 2.6) the mechanism is pure rest-phase dephasing without force or momentum diffusion (§ 8). The experimental consequence is that non-interferometric bounds on Diósi–Penrose (the underground search for spontaneous radiation [14], the Majorana Demonstrator limit [29], and X-ray tests [30]) do not apply. An atomic clock has no mc^2 channel: a superposition of internal levels proceeds with $E_{\rm br} = \Delta E$, and for a microwave

transition χ is negligible. Matter-wave interferometry of the centre of mass is therefore the most direct laboratory probe of the mechanism with $E_{\rm br} = mc^2$.

For the same reason, purely optical interferometers provide no direct constraint ($\mathrm{d}\tau = 0$ for a photon); LIGO and the Holometer constrain external metric and spatial models [31], but not the proposed model. Massive mirrors would be subject to the mechanism only in their own centre-of-mass superposition; in the usual classical optical regime they are not.

7.2 Setup parameters: sodium nanoparticles (Vienna group, 2026).

We confront Equation (7) with the sodium-nanoparticle interferometer of the Vienna group [32, 33]:

- clusters of 5000–10000 Na atoms with a mass-distribution centre $m \approx 172$ kDa ($m \approx 2.86 \times 10^{-22}$ kg);
- velocity $v \approx 160$ m/s;
- three gratings of period $d = 133$ nm spaced by $L = 0.983$ m;
- a measured fringe visibility $V = 0.10 \pm 0.01$ (second run 0.08 ± 0.01) at 172 kDa, described by the quantum model with a single global factor 0.78 for conventional contrast losses (alignment, gravitational and rotational phase averaging, vibrations, thermal and collisional decoherence);
- a record macroscopicity $\mu = 15.5$ [34, 32].

The relevant superposition time is the Talbot flight of one section $t = L/v \approx 6.1$ ms (justification—in § 7.3).

The two resolved branches of a rigid cluster are centre-of-mass shifts of the whole body, with the same internal configuration. At full resolution $\mathcal{D}_{\rm Na} = N$, and Equation (1) gives $E_{\rm br} = N m_{\rm at} c^2 = Mc^2$, so $\chi \propto M^2$. At incomplete resolution the formulas below use $m_{\rm at}\mathcal{D}_{\rm Na}$ in place of M. Full distinguishability of the branches is an explicit assumption and can be checked from the packet geometry.

7.3 Numerical estimate and constraint on ξ.

The superposition time is fixed by the definition of a branch: branch clocks run while the terms of the split are distinguishable. In the interferometer such diffraction orders form at the second grating and recombine at the third, so $t = L/v \approx 6.1$ ms. In this experiment the third grating is a scanned mask of the spatial density, and the published visibility is a Fourier coefficient of the transmitted flux, averaged over masses and velocities. The factor $e^{-\chi}$ equals the ratio of these visibilities if the dephased contribution is unmodulated at that harmonic and a single q at the mean m and v can be taken outside the average. The estimate of ξ below uses this identification. With $\omega_C = mc^2/\hbar = 2.43 \times 10^{29}$ s^{-1} (at $m = 172$ kDa) and $t_P^2 = 2.91 \times 10^{-87}$ s^2, Equation (7) gives

$$\chi \approx 0.41\,\xi^{2/3} \quad (t = L/v \approx 6.1 \text{ ms}). \tag{21}$$

At $\xi = 1$ this is $\chi \approx 0.41$, i.e. a visibility deficit of $\approx 34\%$.

We use the published raw data and analysis code [35]. In the bin $P_2 = 15.252$ mW three independent scans give

$$V_i = 0.0732 \pm 0.0081, \quad 0.0975 \pm 0.0070, \quad 0.0780 \pm 0.0074, \tag{22}$$

where the errors are 1σ from the covariance matrix of a nonlinear sinusoidal fit. For the displayed rounded values, the inverse-variance mean is $\bar{V} = 0.0840$ with a formal error of 0.0043. For

these three scans $\chi^2_{\rm red} = 3.08$; following the experiment's procedure for overdispersed bins, we inflate the error by $\sqrt{\chi^2_{\rm red}}$ and obtain $\sigma_{\bar V} = 0.0076$.

The official quantum curve at this power gives $V_{\rm QM} = 0.1108$ (without conventional losses); it depends on the polarizability, the grating power, and the velocity. The lower bound on the observed visibility at the one-sided normal quantile is

$$\bar V - 1.645\,\sigma_{\bar V} = 0.0716. \tag{23}$$

We construct the upper bound on ξ at $R_{\rm ord} = 1$. The observed visibility decomposes as $V_{\rm obs} = R_{\rm ord}\, V_{\rm QM}\, e^{-\chi}$, where $R_{\rm ord} \le 1$ is the total conventional contrast-loss factor (alignment, vibrations, phase averaging, gas, radiation). Since $R_{\rm ord}$ cannot exceed unity by its physical meaning, regardless of its true value we have $e^{-\chi} \ge V_{\rm obs}/V_{\rm QM}$. Attributing *the entire* visibility deficit to our mechanism ($R_{\rm ord} = 1$), we obtain

$$e^{-\chi} \ge \frac{0.0716}{0.1108} = 0.646, \qquad \chi \le 0.44, \tag{24}$$

and from $\chi = 0.41\,\xi^{2/3}$ we obtain

$$\boxed{\xi \lesssim 1.1.} \tag{25}$$

This bound does not depend on the conventional-loss factor $R_{\rm ord}$: any real losses ($R_{\rm ord} < 1$) could only tighten it. The number 1.1 corresponds to the quantile 1.645 after inflating the error on three scans, that is, two degrees of freedom. Student's quantile $t_{0.95}(2) = 2.92$ gives $\xi \lesssim 1.7$.

If one trusts the published estimate of conventional losses $R_{\rm ord} = 0.78$, the limit tightens: $e^{-\chi} \ge 0.0716/(0.78 \cdot 0.1108) = 0.828$, $\chi \le 0.188$, whence $\xi \lesssim 0.32$. But this is a conditional result: at a single mass the product $R_{\rm ord}\, e^{-\chi}$ is degenerate, and the data do not allow separating the new mechanism from underestimated conventional losses; the interpretation of δt and the overall coefficient ξ also remain model systematics.

Summary for the data: at $\xi = 1$ the model predicts ($R_{\rm ord} = 1$) visibility $0.1108\, e^{-0.41} \approx 0.074$ against the observed 0.0840 ± 0.0076—a deviation of only 1.3σ, so $\xi = \mathcal{O}(1)$ is *not excluded* by the data; the normal quantile gives $\xi \lesssim 1.1$, and the conditional result (at $R_{\rm ord} = 0.78$) gives 0.32. A statistically rigorous limit requires a joint likelihood analysis of the raw scans, the quantum-curve parameters, and all conventional contrast-loss channels.

7.4 The decisive near-term test.

The already-measured high visibility $V = 0.66 \pm 0.09$ at 0.4–1 MDa [32] does *not* test the mechanism as distinguishable quantum coherence: it was taken in the regime $\lambda_{dB} \lesssim 3$ fm, $L \le L_T$ ($L_T \sim d^2/\lambda_{dB}$—the *Talbot length*, the scale of grating self-imaging in the near field of a Talbot–Lau interferometer [32, 22]), where the quantum and classical predictions coincide (geometric optics). A discriminating test requires the quantum regime at high mass. The same programme targets $m \sim 1$ MDa with $v \approx 25$ m/s [32]. Then at $t = L/v \approx 39$ ms and $\omega_{\rm eff} \approx 1.4 \times 10^{30}$ s^{-1}, Equation (7) at $\xi = 1$ gives

$$\chi \approx 47, \qquad V/V_0 = e^{-\chi} \lesssim 10^{-20}. \tag{26}$$

Practically complete loss of interference is robust against $\mathcal{O}(1)$ uncertainty in the coefficient. The standard quantum model of the setup at the same parameters expects measurable fringes with ordinary contrast. Observing fringes at $\sim$1 MDa and $v \approx 25$ m/s would falsify the region $\xi = \mathcal{O}(1)$.

7.5 Distinguishing signature.

A visibility deficit alone is not diagnostic, since thermal radiation, collisions with residual gas, and velocity spread also reduce the contrast. The signature of the present mechanism is its scaling $\chi \propto m^2 t^{2/3}$ together with *immunity to shielding*: it cannot be reduced by improving the vacuum or cooling the particle. The experimental discriminator is whether a residual deficit persists after the suppression of all known environmental channels and whether it follows the m^2 law. Disappearance of fringes at ~ 1 MDa and $v \approx 25$ m/s with scaling $V \propto \exp[-(mc^2/\hbar)^2(\xi t_P^2 t)^{2/3}]$ after excluding environmental losses would be evidence in favour of the model.

An additional discriminator is the dependence on the maximum spatial branch separation Δx. For momentum-transfer decoherence (collisions and photon emission or scattering), the localization exponent contains factors of the form $1 - \mathrm{sinc}(q\Delta x/\hbar)$: it vanishes for unresolved paths, grows as the environment starts to resolve them, and saturates at the event rate once Δx exceeds the environment's spatial resolution [36]. It would therefore be incorrect to say that *all* environmental channels grow without bound with Δx. A gravitational gradient usually produces a deterministic geometry-dependent phase; contrast is lost only after averaging over fluctuations of the gradient or trajectories.

By § 2.4, $\mathcal{D}_s$ depends on the packet separation. While the packets overlap, inserting $\mathcal{D}_s(\Delta x)$ into the exponent is a conditional continuation of the proposed pairwise rule, not a consequence of the map (10):

$$\chi_{\mathrm{BDPT}}(\Delta x) = \left[\frac{c^2 \sum_s m_s \mathcal{D}_s(\Delta x)}{\hbar}\right]^2 (\xi t_P^2 t)^{2/3}. \tag{27}$$

In this continuation $\mathcal{D}_s(\Delta x)$ grows from zero while the packets overlap. Once the branches are resolved, $\mathcal{D}_s = N_s$, the map (10) gives a plateau: a further increase of Δx does not change the energy Mc^2, and χ depends only on m and t.

The test can be performed in one apparatus: at fixed m, v, superposition time, pressure, and temperature, vary the transverse momentum transfer (preferably the diffraction order; changing the grating period also requires compensating the change in Talbot length) and thereby scan Δx. The quantity to compare is not raw visibility but the residual exponent

$$\chi_{\mathrm{res}}(\Delta x) = -\ln\left[\frac{V_{\mathrm{obs}}(\Delta x)}{V_{\mathrm{QM}}(\Delta x)}\right] - \chi_{\mathrm{env}}^{\mathrm{calc}}(\Delta x), \tag{28}$$

where V_{QM} includes coherent dynamics and grating interactions, while $\chi_{\mathrm{env}}^{\mathrm{calc}}$ contains the calculated gas and thermal contributions. The model predicts a constant addition to χ_{res} on the distinguishability plateau, whereas unsaturated environmental channels retain a characteristic Δx dependence. A flat residual alone is not unique: saturated environmental decoherence and Δx-independent technical losses can mimic it. The Δx scan should therefore be combined with changes in vacuum or temperature and with a test of the $m^2 t^{2/3}$ scaling.

8 Distinction from other decoherence mechanisms

The mechanism must be distinguished from nearby sources of interference loss. The closest formal relative is Milburn's intrinsic decoherence [11]; Adler's simplest equation coincides with it [12, 13]: stochastic unitary phase steps generated by the Hamiltonian yield fundamental dephasing specifically in the energy eigenbasis of H, with populations conserved and a rate that grows with $|E - E'|$. BDPT shares with that programme the idea of an internal (not environment-induced) loss of coherence from uncertain time evolution, but is not reducible to it. In Milburn's model decoherence is tied to the eigenstates of the full H and for a free particle produces anomalous dispersion (an effective channel with an operator $\propto p^2$); at $E = E'$

there is no dephasing, so equal-rest-energy which-path arms alone yield no signal. The same vanishing at equal energies, and the factor $t_P^{4/3}t^{2/3}$ in front of the square of a frequency, were already obtained by Gambini, Porto and Pullin from the Salecker–Wigner and Ng–van Dam clock bound: the off-diagonal element decays as $\exp(-\omega_{nm}^2 t_P^{4/3} t^{2/3})$, where ω_{nm} is the difference of eigenvalues of a single Hamiltonian [9, 10]. In Equation (7), $mc^2/\hbar$ stands in place of ω_{nm} by the definitions and the postulate; the numerical coefficient differs by the factor $\xi^{2/3}$. In BDPT the noise attaches to discrete branches by the definitions, and δt is set by the Károlyházy relation with $\chi \propto m^2 t^{2/3}$, and the postulate of independent clocks turn the common Compton phase into a random *relative* contribution without force or momentum diffusion.

The conceptual framework of "proper time along each branch" is already used in quantum-clock interference. In Zych et al., an internal state acts as a clock: a deterministic proper-time difference along the two paths records which-path information in the internal degree of freedom and reduces visibility [15]. Pikovski et al. developed this mechanism for a composite object in a gravitational field: an externally imposed potential difference unitarily entangles the centre of mass with the internal-energy distribution, and dephasing appears after tracing out the internal degrees of freedom [16]. In contrast, the present model does not rely on an external potential difference or on a dedicated internal degree of freedom that unitarily encodes which-path information; it adopts the Károlyházy relation for δt and the postulate of branch-clock independence, and therefore predicts irreversible ensemble dephasing even for an object in its internal ground state.

An environment with momentum transfer (gas, thermal radiation, photon scattering) creates position-dependent noise, force, and momentum diffusion and is usually suppressed by improving the vacuum or cooling; the characteristic dependence on Δx is described by localization factors [36] (see § 7.5). The closest construction of this kind is Diósi's local-time uncertainty [13]: the rest energy $c^2 f(r)$ couples to random shifts of local time with correlator $\propto G\hbar/(|r-r'|c^4)$. With that correlator c cancels, and the master equation coincides with the Diósi–Penrose equation, whose rate is set by the gravitational self-energy. In the present model the shifts are independent across branches and their variance is fixed by the relation (3), so c remains and the rate is set by mc^2. Dynamical Diósi–Penrose collapse models predict population redistribution and spontaneous emission [14], to which the present model is *transparent*: it is pure rest-phase dephasing without additional channels. Spatially attached implementations of Planckian uncertainty (correlated shear noise, external metric fluctuations) are constrained by devices such as the Holometer and LIGO-like detectors [31]; BDPT attaches to discrete branches, not to points, and does not act on a massless carrier. Finally, in standard quantum mechanics the Compton phase cancels in a closed interferometer [23]; the present model does not settle that debate but postulates that, with independent branch clocks, the rest phase mc^2 becomes a random *relative* contribution (see § 3)—a separate and falsifiable premise.

The Margolus–Levitin theorem gives the orthogonal-evolution rate $\nu_{\max} = 2E_{\mathrm{av}}/(\pi\hbar)$, where E_{av} is the energy above the ground level [37]. The additional identification $E_{\mathrm{av}} = mc^2$, which does not follow from the definitions and the postulate, gives the formal rate $N = (2/\pi)\, t/t_C$. Combining it with the coherence condition $t_{\mathrm{dec}} \geq t$, i.e. $t_C \geq (\xi\, t_P^2\, t)^{1/3}$, and taking equality at the boundary gives

$$N_{\mathrm{formal}}(t) = \frac{2}{\pi\, \xi^{1/3}} \left(\frac{t}{t_P}\right)^{2/3} . \tag{29}$$

Over one second this is $\approx 4.5 \times 10^{28}$ at $\xi = 1$. The exponent 2/3 coincides with the scaling $N \propto (t/t_P)^{2/3}$ obtained by Ng for a "space-time-foam computer" and a black-hole quantum computer [18], because both estimates use the cubic law $\delta t^3 \sim t_P^2 t$. This is a meaningful structural analogy, but not an independent derivation of an operation-count bound: without additional physical justification for $E_{\mathrm{av}} = mc^2$, matching exponents does not turn N_{formal} into a computational limit of the model.

9 Open questions

The open questions for the further development of the model are collected here.

9.1 Branch basis.

The clocks belong to the split whose interference is read out. On a given interval that split is fixed by the preparation, the interferometer, or, for internal levels, the eigenbasis of H_{int}, and this choice is held until the interval ends. Another writing of the same vector does not add clocks. In a qubit register the variance of a fixed pair does not depend on the number of the other branches, but pairs with different branch energies have different variances. A change of basis defines a different split, and the full-resolution map does not generate it. The model has no general rule for a superposition that has neither a prepared which-path split nor a distinguished H_{int}. A NOON state with one clock for the collective branch and a product of one-atom splits are two different physical conditions, not two writings of one state.

9.2 Partner-dependent energy.

The equality $E_{a|b} = E_{b|a}$ does not make the noise consistent for three or more branches when the energy of a branch depends on which other branch it is compared with: the relative phases may then fail to close a cycle. When packets overlap only in part, that energy is different for each pair, the packets are not given as orthogonal projectors, and there is no single overall map. For a mode displaced by much less than the atomic zero-point motion an orthogonal basis is likewise not given: the energy that enters is the mode energy rather than a pairwise particle count, and there is no single joint noise with one overall map. The point is that a consistent map needs one energy and one phase per branch; a partner-dependent pair energy may fail to supply that set of phases, and the averaged matrix may then lose positivity. A possible direction is to assign one energy to each branch of an already given orthogonal decomposition and to average the diagonal unitaries. While the packets overlap, $\mathcal{D}_s$ remains an estimate of the fraction of that energy. For a soft mode the basis is the states of the mode, and the energy is the mode energy.

9.3 Repeated splittings.

The map derived in § 3.2 covers one interval while the arms propagate separately. There is no rule for composing repeated splittings and recombinations. The point is that the map of one interval does not say what happens when the arms merge and when they split again: the clocks may stop or keep a memory, and the physical applicability of composing the maps without accounting for the memory of the noise has not been established. A possible direction is, on merging, to pass the relative phase into the interference and to stop the clocks of that split, and to start a new splitting with new independent shifts on a new interval.

9.4 Spatial correlation of the noise.

The assumption in § 2.6 fixes only a threshold: the correlation of the branch noises may be appreciable for a separation smaller than $\ell = c\delta t$ and is absent at a larger separation. The precise dependence of the correlation on distance does not follow from the relation (3) and is not derived in this work. Until it is, resolved arms of a laboratory interferometer are assigned independent shifts.

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